Project_2_Customer_Churn_Analysis (2) (1)

February 16, 2026

step 1 load data

```
[1]: import pandas as pd
     import numpy as np
     df=pd.read_csv("/customer_churn_synthetic.csv")
     print(df.head())
```

step 2 Churn Rate

```
[ ]: churn_rate =  df["churn"].mean()*100
     print("Churn Rate":,churn_rate)
```

step 3 Visualization

```
[ ]: import matplotlib.pyplot as plt
     df["churn"].value_counts().plot(kind='pie',autopct='%1.1f%%')
     plt.show()
```

step 4 Business Analysis

Tenure Vs churn

```
[ ]: df.groupby("churn")
     ["Tenure__Months"].mean()
```

Balance vs churn

```
[ ]: df.groupby("churn")
     ["Balance"].mean()
```

#Active vs churn

```
[ ]: pd.crosstab(df["IsActiveMember"],df['Churn'])
```

step 5 Machine Learning Model

Features & Target

```
[ ]: X = df.drop(["CustomerID,"Churn",
     'Gender'],axis=1)
     Y = df["Churn"]
```

Train test split

```
[ ]: from sklearn.model_selection
     import train_test_split
     X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2)
```

```
[ ]: from sklearn.linear_model
     import pandas as pd
     model=LogisticRegression()
     model.fit(X_train,y_train)
     print("Accuracy:",model.score(X_test,y_test))
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```